

Problem definition

Although recruitment platforms have been fully digitised, efficiency has not really improved. Instead, they were stuck with "data overload but matching failure." According to current market observation, although the algorithm has begun to intervene, the user experience is still poor. This problem can be mainly broken down into the following core technologies and trust bottlenecks:

First of all, the existing matching mechanism is too superficial and lacks "semantic understanding". Most platforms are still limited to simple keyword matching. This leads to a serious problem: job seekers submit numerous applications only to disappear without a trace, while employers are inundated with irrelevant or unqualified resumes. This keyword-based filtering method fails to identify the depth of skills in a resume or the true meaning of the job requirements, resulting in a severe skills mismatch.

Secondly, the existing system lacks the logic of bidirectional optimisation. Current hiring processes are often one-sided—either candidates are searching aimlessly, or HR professionals are struggling to conduct exhaustive screening. True recruitment should be a two-way, precise match; however, current algorithms rarely balance the dynamic needs and preferences of both parties at the same time. This directly leads to the current situation of a low interview conversion rate and extremely high communication costs.

Additionally, third-party involvement can also lead to a crisis of trust and pose privacy risks. In the current recruitment ecosystem, too many third-party intermediaries not only fragment the information delivery but also reduce the transparency of the system. More critically, job seekers' sensitive information (such as contact information, salary expectations and work experience) often faces significant privacy risks on these platforms. When job seekers lose trust in the platform, high-quality talent will be lost, further weakening the effectiveness of recruitment, and the goodwill of the platform will also decline.

To sum up, what is currently missing is not a "place to post job vacancies", but an intelligent system that can truly understand resumes and job requirements and perform automated two-way matching while ensuring data security. This is exactly the core challenge that this project aims to solve: by developing an "intelligent talent matching platform" to transform inefficient manual screening into high-precision algorithm matching.